INCH-POUND

MIL-N-53097(ME)
7 February 1990

MILITARY SPECIFICATION

NOZZLE, PRESSURE FUEL SERVICING, LOCKING, TYPE D-1, NOMINAL 2-1/2-INCH DIAMETER, ARCTIC SERVICE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document specifies the requirements for a pressure fuel servicing nozzle, designated type D-1, which has a 45-degree inlet with a nominal diameter of 2-1/2-inches and is used for dispensing liquid fuel in a low temperature (-60 °F) environment to aircraft and ground equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-C-1752	- Cushioning Material, Packaging, Uncellular Polyethylene Foam, Flexible.
PPP-C-1842	- Cushioning Material, Plastic, Open Cell (for Packaging Applications).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-P-5315	- Packing, Preformed, Hydrocarbon Fuel Resistant.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum Alloys.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-7024	- Calibrating Fluid, Aircraft Fuel System Components.
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series, General Specification for.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-S-8879	- Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for.
MIL-R-25988	- Rubber, Flurosilicone Elastomer, Oil and Fuel Resistant Sheets, Strips, Molded Parts, and Extruded Shapes.
MIL-T-38219	- Turbine Fuel, Low Volatility, JP-7.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
MIL-R-83248	- Rubber, Fluorocarbon Elastomer, High Temperature Fluid and Compression Set Resistant.
MIL-C-83413	- Connectors and Assemblies, Electrical, Aircraft Grounding, General Specification for.
MIL-C-83413/7	- Connectors and Assemblies, Electrical, Aircraft Grounding: Grounding Clamp Connector for Types I and III Grounding Assemblies, Clip, Electrical.
MIL-C-83488	- Coating, Aluminum Vapor Deposited.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking for U.S. Military Property.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-882	- System Safety Program Requirements.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1523	- Age Control of Age Sensitive Elastomeric Material.
MIL-STD-2175	- Casting, Classification and Inspection of.
MS 24484	- Adapter, Pressure Fuel Servicing, Aircraft, Nominal 2-1/2 Inch Diameter.
MS 25384	- Electrostatic Discharge Jumper, Fuel Nozzle to Aircraft.
MS 26551	- Strainer, 2-1/2 Inch Pressure Fuel Servicing.
MS 29513	- Packing, Preformed, Hydrocarbon Fuel Resistant, "O" Ring.
MS 29520	- Envelope Dimensions, Nozzle, Pressure Fuel Servicing, Locking, Aircraft, Type D-1 and D-2.
MS 33666	- Packing, Preformed - Aeronautical, Elastomeric, Range of Sizes.

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MS 33786 - Fitting Installation, Flared Tube and Hose, Swivel.

(Unless otherwise indicated, copies of federal and military specifications and standards, are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13228E1820 - Gasket.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 471 - Rubber Property - Effects of Liquids.
D 910 - Aviation Gasolines, Specifications for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The D-1 nozzle shall incorporate a 45-degree inlet configuration with an inlet swivel fitting capable of rotating 360 degrees. The nozzle interface and mating adapter and nozzle operating and control device shall be as specified herein.

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3.1.1 Drawing. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawing to insure correct fit, assembly, and operation of the nozzle. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials used in the manufacture of nozzles shall be resistant to all fluids as specified in TT-S-735, MIL-C-7024, ASTM D 910, MIL-T-5624, MIL-T-38219, and MIL-T-83133. The use of copper, cadmium, magnesium, or any alloy thereof is prohibited.

3.3.1 Material deterioration prevention and control. The nozzles shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the nozzles may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the nozzles may be newly fabricated from recovered materials to the maximum extent practicable, provided the nozzles produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the nozzle.

3.3.3 Metals. Corrosion-resistant metals shall be used. Exterior surfaces susceptible to contact with ferrous surfaces during use shall be nonferrous to restrict spark generation.

3.3.3.1 Protective coatings. All aluminum alloy parts shall be protected by coatings in accordance with MIL-A-8625, MIL-C-5541, MIL-C-83488, or other protective coatings which will withstand the effects of this specification.

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3.3.4 Castings. Castings shall be classified and inspected as specified in MIL-STD-2175.

3.4 Design. The nozzle shall be designed to operate within the clearance envelope described in MS29520 and shall mate adapter conforming to MS24484 within a misalignment angle up to 4 degrees. The nozzle shall transfer fuel at flows up to 800 gallons per minute (gpm) with operating pressures of up to 55 psi (pounds per square inch). The fuel flow shall be controlled by an ON/OFF fuel flow actuation lever separate from the handles used to hold or attach the nozzle to the adapter. Actuation of the lever shall simultaneously open or close the adapter sealing unit and permit flow to or from the aircraft and ground equipment. Operation of the lever from the nonflow to flow position shall be impossible unless the nozzle is secured onto the adapter. Removal of the nozzle from the adapter shall be impossible unless the lever is in the nonflow position. The lever shall remain in either the full flow or nonflow position under the full range of flows and pressures, during fueling or defueling, without the need of maintaining continuous manual holding force. To prevent leakage, the nozzle-to-adapter main seal shall automatically conform to the adapter when the nozzle is attached and opened. The seal must compensate for 0.020-inch wear on the adapter lugs, dimensional tolerances, side loading, and temperature changes. Provision should be made for the installation and retention of a screen conforming to MS 26551. The inlet flange shall be designed to accommodate a gasket conforming to Drawing 13228E1820.

3.4.1 Swivel inlet. The nozzle shall employ a swivel inlet fitting capable of 360-degree rotation to allow the hose connection to turn without hinderance.

3.4.2 Dust cap. When tested in accordance with MIL-STD-810 (see 4.5.2.7), nonmetallic cap shall be provided to prevent dust from entering that portion of the nozzle which engages the adapter and protect the seal face of the nozzle during handling. When not in use, the cap shall hang free outside the refueling envelope so that it will not interfere with the refueling. Retention springs are not permitted.

3.4.3 Lubricants. The nozzle shall operate satisfactorily without the use of lubricants.

3.4.4 "O" ring packing and seals. "O" ring packing and associated seals shall meet the operating (see 3.11) and storage (see 3.12) temperature requirements and shall conform to MIL-P-5315, MIL-R-25988, MIL-R-83248, MS 29513, or MS 33666 as applicable.

3.4.4.1 Age controls for synthetic rubber parts. Age sensitive synthetic components shall be marked in accordance with MIL-STD-1523.

3.4.5 Maintenance. The nozzle shall not require special tools or fixtures for overhaul or repair of the assembly or any of its components.

3.4.6 Pressure drop. The pressure drop through the nozzle and the adapter confirming to MS 24487 shall be not greater than 10 (psi) corrected when subjected to a 600 gpm flow of fuel with no strainer conforming to MS 26551 installed. In the defueling mode, the pressure drop through the nozzle and adapter

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shall be not greater than 2 psi corrected when subjected to a 200 gpm flow of fuel with the strainer removed. The test setup shall be in accordance with figure 1.

3.4.7 Flow-actuation lever. The flow-actuation lever shall be capable of operating from full closed to full open with a maximum torque of 40 pounds force inch (lbf.in) when a force of 50 pounds is applied against the nozzle poppet in the direction to close the nozzle. The lever shall be of adequate size to be grasped easily by personnel wearing arctic mittens. The lever shall be provided with a double-ended arrow showing direction of operation, labeled at each end to indicate the functional result.

3.4.8 Strength. When tested in accordance with 4.5.2.9, the nozzle shall be able to withstand the loads shown on figures 2 and 3 without leakage, distortion, or failure.

3.4.9 Fuel quality sampling port. The nozzle shall be provided with a sampling port boss having a .375-inch National Pipe Thread (NPT). Location of the boss shall allow insertion of a sampling probe which can penetrate 0.75-inch within the nozzle flow stream. Insertion of the probe shall not interfere with the operation of the nozzle or the screen confirming to MS 26551.

3.4.10 Electrical continuity. Electrical continuity between the nozzle and adapter shall be established through the attaching lugs when the units are connected. The resistance from the nozzle inlet flange to the adapter locking flange shall be not greater than 10 ohms.

3.4.11 Human factors engineering. The nozzle shall conform to Human Factors Engineering Design Criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, 4 (General Requirements), 5.9 (Design for Maintainability) and 5.13 (Hazards and Safety) of MIL-STD-1472 as applicable.

3.4.12 Ground cable assembly. The nozzle assembly shall be equipped with a ground cable assembly for grounding the nozzle to the vehicle and to a ground stake. The nozzle-to-vehicle ground cable shall conform to MS 25384-1, except the connector shall be coated wire 15 feet long, and the end shall be equipped with a clamp in accordance with MIL-C-83413/7, suitable for bonding the cable assembly to a 5/8-inch diameter ground stake.

3.5 Performance. When subjected to any tests of 4.5.2, in the order listed, the nozzle shall satisfy all performance requirements.

3.6 Weight. The weight of the nozzle shall be not greater than 9 pounds.

3.7 Threads. All machine screw threads shall be in accordance with MIL-S-7742 or MIL-S-8879.

3.7.1 Locking of threaded parts. All threaded parts shall be locked to prevent loosening or detachment which would result in the part entering the system being refueled or cause foreign object damage to the aircraft and ground equipment.

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3.8 Operational/maintenance manual. A manual which provides specific instruction for the operation, maintenance, and repair of the nozzle by refueling personnel shall be provided (see 4.5, 5.1.1, and 5.1.2).

3.9 Identification of nozzle. The nozzle shall be marked for identification in accordance with MIL-STD-130. The identification shall be permanent and its location shall not affect nozzle performance. The minimum information is as follows:

Nozzle, Pressure Fuel Servicing, Arctic Service
Contract No.
Type D-1
Manufacturer's identification, part number, serial number
Cure date per MIL-STD-1523 (if applicable)

3.10 Safety. The nozzle shall be designed to minimize fuel sprays or spills resulting from failure of any nozzle component.

3.11 Operating temperature. The nozzle shall perform as specified herein at all ambient temperatures between -60 and +95 °F.

3.12 Storage temperature. The nozzle shall not be damaged by storage at all ambient temperatures between -65 and +145 °F.

3.13 Workmanship. The nozzle shall be uniform in quality, clean and free from faults and burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Disassembly inspection. Failure of any examination or test of the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawing in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).
d. Inspection comparison (see 4.7).

4.3 First article inspection. Unless otherwise specified (see 6.2), two first article nozzle assemblies shall be examined and tested as specified in 4.5.1 and 4.5.2. Presence of one or more defects or failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), the Government reserves the right to perform first article testing on three or more nozzles. The tests shall consist of any of the tests of 4.5.2 which the Government desires to perform.

4.3.1 Rejection and retest. When a nozzle fails to meet any of the requirements of the first article tests, the cause of failure shall be determined. Full particulars of the failure and the action required to correct the fault shall be reported to the contracting officer. All first article tests which are affected by the corrections shall be repeated.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examinations and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each nozzle shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

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4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The nozzle shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration in the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment.
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the nozzle assembly.
106. Castings not classified and inspected as specified.
107. Age sensitive synthetic components not marked as specified.
108. Human factors engineering not as specified.
109. Weight of the nozzle not as specified.
110. Threads not as specified.
111. Operational/maintenance manual missing or not as specified.
112. Identification of nozzle missing or not as specified.
113. Safety not as specified.
114. Operating temperature not as specified.
115. Storage temperature not as specified.
116. Workmanship not as specified.
117. Ground cable assembly not as specified.

4.5.2 Tests.

4.5.2.1 Test schedule.

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TABLE I. Test schedule.

First article	Quality conformance		Test	Test paragraph	Requirement paragraph
	Individual	Samples			
1	2	3	4	5	6
X	X	X	Test conditions.	4.5.2.2	3.5
			Calibration.	-	-
X	X	X	Functional.	4.5.2.3.1	3.5
X		X	Pressure drop.	4.5.2.3.2	3.5
			Leakage.	-	-
X	X		Connected.	4.5.2.4.1	3.5
X			Disconnected.	4.5.2.4.2	3.5
X			High flow shut-off.	4.5.2.5	3.5
X			Fuel resistance and low temperature.	4.5.2.6	3.5
X			Dust.	4.5.2.7	3.5
X			Swivel endurance.	4.5.2.8	3.5
X			Strength.	4.5.2.9	3.5
X			Endurance.	4.5.2.10	3.5
			Normal operation.	4.5.2.10.1	3.5
X			Maximum loading.	4.5.2.10.2	3.5
X			Salt water immersion.	4.5.2.11	3.5
X		X	Proof pressure.	4.5.2.12	3.5
X			Burst pressure.	4.5.2.13	3.5
X			Drop test.	4.5.2.14	3.5
X			Electrical conductivity.	4.5.2.15	3.5
X		X	Disassembly and inspection.	4.5.2.16	3.5

4.5.2.2 Test conditions. Unless otherwise specified (see 6.2), the following test conditions shall apply.

4.5.2.2.1 Test fluid. Fluid in accordance with ASTM D 910; MIL-T-5624, grades JP-4 or JP-5; MIL-T-38219; or MIL-T-83133, grade JP-8 shall be used for testing. When a specific fluid is not specified (see 6.2), fluid conforming to MIL-C-7024, type II, or any fluid acceptable to the contracting activity, may be used. Test fluid should be at -60 °F for testing at -60 °F.

4.5.2.2.2 Ambient temperature and pressure. All tests shall be performed with the ambient and test fluid at temperatures of -60 to +95 °F and at local barometric pressure.

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4.5.2.2.3 Test adapter. The test adapter used shall be in accordance with ASTM D 471, Reference Fuel A and B, and MS24484, installed in a test housing (see figure 4).

4.5.2.3 Calibration.

4.5.2.3.1 Functional. The nozzle shall be inserted and attached onto the test adapter (see figure 4) and the adapter sealing valve actuated a minimum of five times. Test fluid shall be passed through the nozzle and adapter with flows from 100 to 800 gpm in 100-gpm increments with a minimum inlet static pressure of 40 psig (pounds per square inch gauge) under flow conditions and 55 psig in the nonflow position. The adapter sealing valve shall be actuated to open and shut a minimum of three times at each flow condition, and there shall be no evidence of binding or chattering of the lever. The torque required to actuate the lever shall be less than 40 lbf.in. under no-flow conditions.

4.5.2.3.2 Pressure drop. Using a test setup similar to figure 1, and the adapter and test housing in accordance with figure 4, a pressure drop test shall be performed with flows from 0 to 800 gpm, in 100-gpm increments, with an inlet pressure of 40 psi. Test results shall reflect both actual and corrected pressure drops. The formula for correction is as follows:

$$AP = AP_{NA} \times \frac{1.34}{m^{.25}_{cs}} \times \frac{.769}{r}$$

where

m = kinematic viscosity of test fluid
r = specific gravity of test fluid
AP_{NA} = total pressure drop - pressure drop due to piping

4.5.2.4 Leakage.

4.5.2.4.1 Connected. With the nozzle connected to the adapter (see figure 4) and the nozzle flow-actuation lever in the full open position, the leakage test shall be conducted over a range of 12 to 4 psig in 2-psi increments and from 10 to 60 psig in 10-psi increments. The pressure shall be maintained at each increment for at least one minute. There shall be no evidence of external fuel leakage from the nozzle, and air leakage inward into the nozzle shall not exceed 50 cc (cubic centimeter) per minute during any part of the leakage test.

4.5.2.4.2 Disconnected. With the nozzle disconnected from the adapter, fuel pressures of 0 to 60 psig in 10-psi increments, air at pressures from -12 to 0 psig in 2-psi increments, shall be applied to the nozzle inlet. Pressure shall be maintained for at least one minute at each increment. There shall be no external fuel leakage nor air leakage into the nozzle.

4.5.2.5 High flow shutoff. With the nozzle connected to the adapter (figure 4) and test housing, conduct ten rapid shutoffs with the flow-actuation lever while flowing 400 gpm and 600 gpm at an inlet pressure of 60 psig. There shall be no evidence of deformation, fracture, or malfunction of the nozzle.

4.5.2.6 Fuel resistance and low temperature. The fuel resistance and low temperature tests shall be conducted in accordance with table II.

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TABLE II. Fuel resistance and low temperature test schedule.

Test Period 1/	Fuel Resistance				Low Temperature
	Phase I, Soak	Phase I, Dry	Phase II, Soak	Phase II, Dry	
Component configuration	Valve closed during first 72 hours and open during last 24 hours 2/	Drained, valve closed, inlet port open	Valve closed during first 12 hours and open during last 6 hours 2/	Drained, valve closed, inlet port open	Valve closed 2/
Test Fluid	ASTM D 471, Reference Fuel B	None	ASTM D 471, Reference Fuel B	None	ASTM D 471, Reference Fuel A
Period duration	96 hours (4 days)	24 hours	18 hours	30 hours	18 hours
Ambient and test fluid temperature	95 ±5 °F	Circulating air at 95 ±5 °F	95 ±5 °F	Circulating air at 95 ±5 °F	Lower the fluid temperature to -65 °F maximum, than maintain the fluid temperature at -65 °F maximum for a period of 18 hours
Operation or tests during period	Connect and disconnect nozzle and actuate valve once every 6 hours	None	Connect and disconnect nozzle and actuate valve once every 6 hours	None	None

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TABLE II. Fuel resistance and low temperature test schedule - Continued

Test Period 1/	Fuel Resistance				Low Temperature
	Phase I, Soak	Phase I, Dry	Phase II, Soak	Phase II, Dry	
Operation or tests immediately after period	Conduct leakage test (4.5.2.4), using ASTM D 471, Reference Fuel A at positive pressures only.	Connect and disconnect nozzle and actuate valve once in dry condition. Conduct leakage test (4.5.2.4), using ASTM D 471, Reference Fuel A at positive pressures only.	Same as phase I, soak	Same as phase I, dry	With ambient tempt not higher than -60 °F: (a) operate the nozzle through 200 complete cycles; conduct test specified in 4.5.2.10.1, (b) conduct leakage test, using ASTM D 471 Reference Fuel A at positive pressures only 3/

1/ Each period shall follow immediately after the preceding one in the order noted.
2/ The component shall be maintained in such a manner as to insure complete contact of all nonmetallic parts with the fluid as would be expected under normal service conditions. An assembly test housing conforming to figure 4 shall be used when the valve is open. No connection shall be made during the soak period when the nozzle is closed, except for test purposes.
3/ For the low temperature test (see 4.5.2.6), the leakage limits are:
Static: 0-4 psig - No leakage
Static: 4-60 psig - 2 ml per minute
During actuation: 0-60 psig - 2 ml per actuation

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4.5.2.7 Dust. The nozzle, with the dust cover in place and the inlet sealed, shall be subjected to test method 510.2 of MIL-STD-810. The functional test of 4.5.2.3.1 shall be conducted following completion of the dust test.

4.5.2.8 Swivel endurance. The nozzle swivel shall be wear-tested by loading the nozzle with a 100 lbf.ft (pounds force foot) bending load (see figure 2) and turning the swivel 2,000 times through at least a 45-degree arc. During this test, the nozzle will be pressurized with test fluid to 55 psig. Following this loading, the nozzle shall successfully complete the proof pressure test of 4.5.2.12.

4.5.2.9 Strength. The adapter shall be retained by the mounting flange only, to a fluid container capable of sustaining any pressures applied. The nozzle shall be connected and the adapter valve actuated to the flow position. The nozzle test adapter shall be filled with test fluid. There shall be no evidence of leakage, deterioration, deformation, fracture, or malfunction when the following loads are applied to the nozzle (figures 2 and 3).

a. 25 applications of a 600 lbf.ft bending load, applied as shown on figure 2.
b. 950 applications of a 350 lbf.ft bending load, applied as shown on figure 2.
c. 25 applications of a 200 lbf.ft bending load, applied as shown on figure 2, with a fluid pressure of 240 psig applied to the adapter-nozzle combination.
d. 100 applications of a combined load of a 100 lbf.ft couple applied to the nozzle handles and an axial load of 200 pounds applied to the nozzle inlet flange in accordance with figure 3, with a fluid pressure of 240 psig applied to the adapter-nozzle combination.

4.5.2.10 Endurance. There shall be no evidence of binding, chatter, or leakage during the tests of 4.5.2.10.1 and 4.5.2.10.2. The force to actuate the lever shall not exceed the requirements of 4.5.2.3.1.

4.5.2.10.1 Normal operation. The nozzle shall be operated for 2,000 cycles (see 6.4.1) with fuel pressure of 55 psig applied to the nozzle in the closed position and with fuel flow through the nozzle in the open position at a minimum rate of 400 gpm. The 2,000 test cycles shall include 200 cycles at each of the following ambient temperatures: -55, -50, -40, -30, -10, +30, +50, +70, and +95 °F.

4.5.2.10.2 Maximum loading. The nozzle, filled with fuel, shall be connected to the adapter in a test setup capable of imposing a 25 +3, -0 pound, load on the poppet in the closed position in a direction tending to close the valve. After the valve is opened 0.75 inch, the load shall be increased to 50 +5, -0 pounds. The valve shall be fully opened and closed 2,000 times. Upon completion of this test, the leakage test (see 4.5.2.4) shall be conducted at a positive pressure only.

4.5.2.11 Salt water immersion. The nozzle shall be immersed in a solution consisting of 5 percent by weight of NaCl in distilled water. After immersion, the solution shall be drained and the nozzle shall be heated in an oven to a

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temperature of 130 ±5 °F for a period of not less than one hour. The immersion and heating cycle shall be repeated 50 times. The nozzle shall not be operated at any time during the immersion cycles. Immediately after completing the immersion cycles, the nozzle shall be washed with warm water to remove all salt accumulations. The nozzle shall be dried; wetted with fuel; and operated for three complete cycles (see 6.4.1). Corrosion of any part of the nozzle to a degree that affects performance shall be cause for rejection.

4.5.2.12 Proof pressure. With the nozzle in the closed position, it shall be subjected to a pressure which shall be not less than 180 psig for at least one minute. There shall be no evidence of leakage, distortion, or other damage to any part of the assembly.

4.5.2.13 Burst pressure. With the nozzle in the closed position, it shall be subjected to a pressure which shall not be less than 240 psig for at least one minute. There shall be no evidence of distortion or other damage to any part of the nozzle; and when the pressure is lowered to 60 psig, there shall be no evidence of leakage from any portion of the nozzle.

4.5.2.14 Drop test. The nozzle, attached to a 2-1/2-inch ground refueling hose, shall be dropped at least ten times from a height of 6 feet onto a steel plate. Drops shall be configured to impact at least once upon the handles, lever, and the nozzle seal face. The remaining impact shall be at random locations as specified by the contracting officer. The steel plate shall be at least 1/2 inch thick, resting on a concrete floor. The upstream portion of the hose shall lie at floor level during the test. Upon completion of this test, the nozzle shall satisfactorily pass the functional (4.5.2.3.1) and leakage (4.5.2.4) tests.

4.5.2.15 Electrical conductivity. With the nozzle attached to the adapter, a resistance reading shall be made between the nozzle inlet and the adapter mounting flange. Resistance shall be not greater than 10 ohms.

4.5.2.16 Disassembly and inspection. The nozzle shall be disassembled for inspection of all parts. There shall be no defects that would cause the nozzles to fail to function within specification limits.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection.

4.6.1.1 Unit of product. The unit of product shall be a quantity of nozzles preserved and packaged in a single container.

4.6.1.2 Sampling. Sampling shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects:

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118. Nozzles, gaskets and manuals not preserved as specified for level A or C (see 5.1.1 and 5.1.2).
119. Shipping containers not as specified for levels A and C (see 5.2.1 and 5.2.2).
120. Shipping container weight and size exceeding specified limitations for levels A and C (see 5.2.1 and 5.2.2).
121. Strapping not as specified for level A (see 5.2.1).
122. Marking incorrect, missing or illegible (see 5.3).

4.7. Inspection comparison. The Government may select nozzles at any time during the contract production period and subject these nozzles to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article nozzles. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison. The nozzle shall be subjected to inspection for adherence to the envelope requirements of MS29520, Drawing 13228E1820 and manufacturer's drawings.

4.7.1 Inspection failure. Failure of any nozzle to meet any requirement specified herein, during, and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the nozzle and shall be cause for refusal by the Government to continue acceptance of production nozzles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on nozzles produced and accepted under the contract. Any deficiencies found as a result of first article inspection will be considered prima facie evidence that all nozzles accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each nozzle shall be preserved in accordance with the method III requirements of MIL-P-116, utilizing a fiberboard carton as the unit container; the carton shall conform with PPP-B-636, class weather resistant, grade W6c. Cushioning material conforming to PPP-C-1752 or PPP-C-1482 shall be used, as necessary. One gasket in accordance with Drawing 13228E1820 and the operational/technical manual shall be individually preserved in accordance with method IC-3 of MIL-P-116. Both shall be included in the unit pack.

5.1.2 Level C. Each nozzle shall be preserved as specified for level A, except the boxes shall be class domestic.

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5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Nozzles, preserved as specified in 5.1 shall be packed in boxes conforming with PPP-B-601, overseas type, style optional. The gross weight of the shipping container and contents shall not exceed 150 pounds. Boxes shall be strapped in accordance with the appendix to the box specification except strapping shall be flat steel and shall be finish B.

5.2.2 Level C. Nozzles, preserved as specified in 5.1, shall be packed as specified for level A, except the boxes shall be domestic type and the strapping may be finish A.

5.3 Marking. Marking of both unit and shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nozzle covered by this specification is intended for use in the refueling or defueling of aircraft and ground equipment by means of a pressure fuel servicing system in a low temperature (-60 °F) environment. The type D-1 nozzle incorporates a 45-degree inlet configuration for preferential horizontal refueling.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issues of individual documents referenced (see 2.1.1 and 2.2).
c. When submission of first article nozzle(s) is not required (see 3.2).
d. Number of first article nozzles when required (see 3.2 and 4.3).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2 and 4.3).
f. When test conditions do not apply (see 4.5.2.2).
g. When a specific test fluid is specified (see 4.5.2.2.1).
h. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must

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furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Cycle. Unless otherwise specified, a cycle is defined as connecting the nozzle to the test adapter; fully opening and closing the flow-actuation lever four times; and removing the nozzle from the test adapter.

6.5 Data requirements. The contracting officer should include requirements for such data as an operational/maintenance manual detailing the recommended operations, maintenance, and repair of the nozzle.

6.6 Subject term (key word) listing.

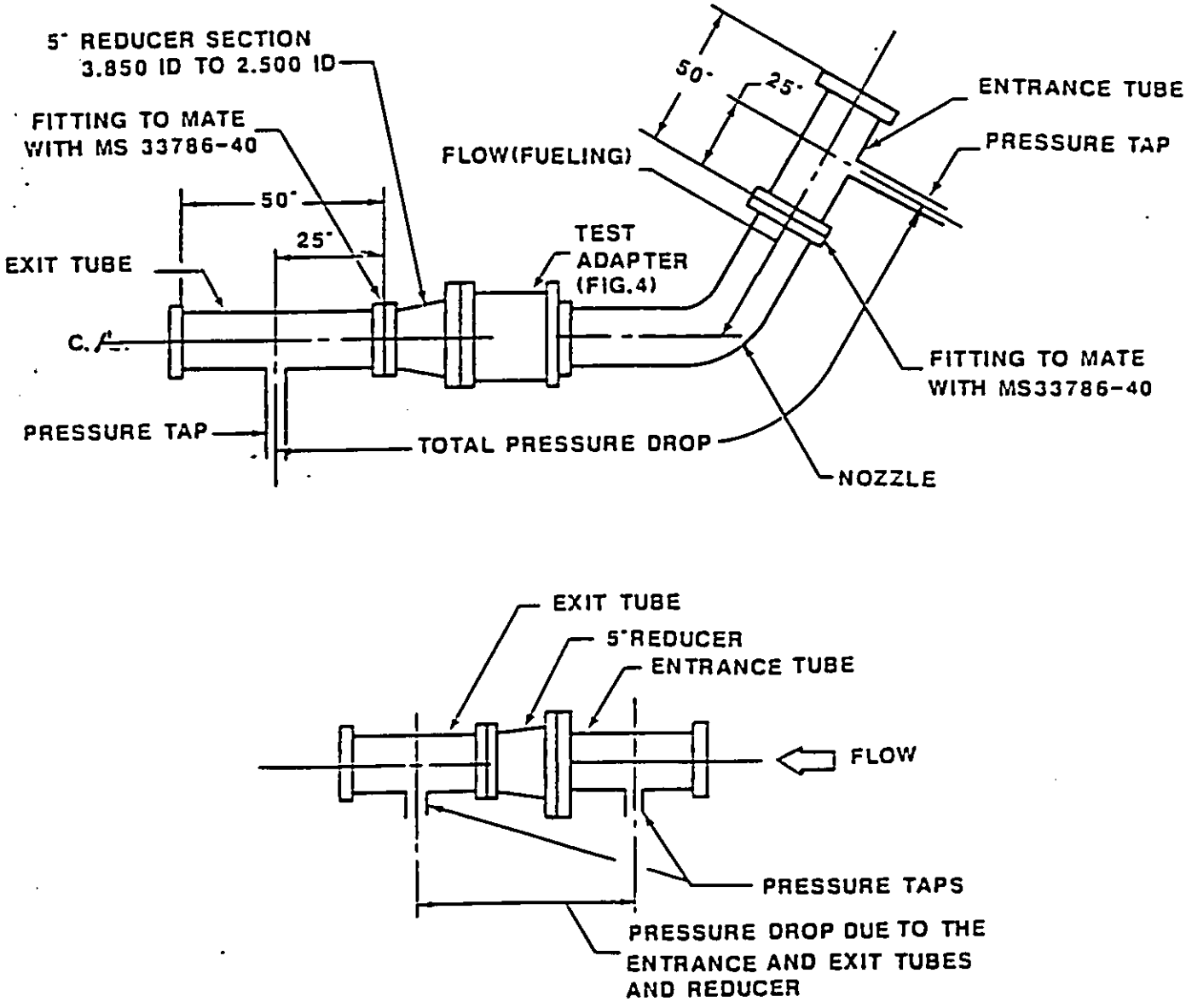
Aircraft
Fueling
Ground equipment
Low temperature
Nozzle
Refueling

Custodian:
Army - ME

Preparing activity:
Army - ME

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NOTE:

Nominal tube size - Entrance and exit tube - 2-1/2" Schedule 40

FIGURE 1. Pressure drop test.

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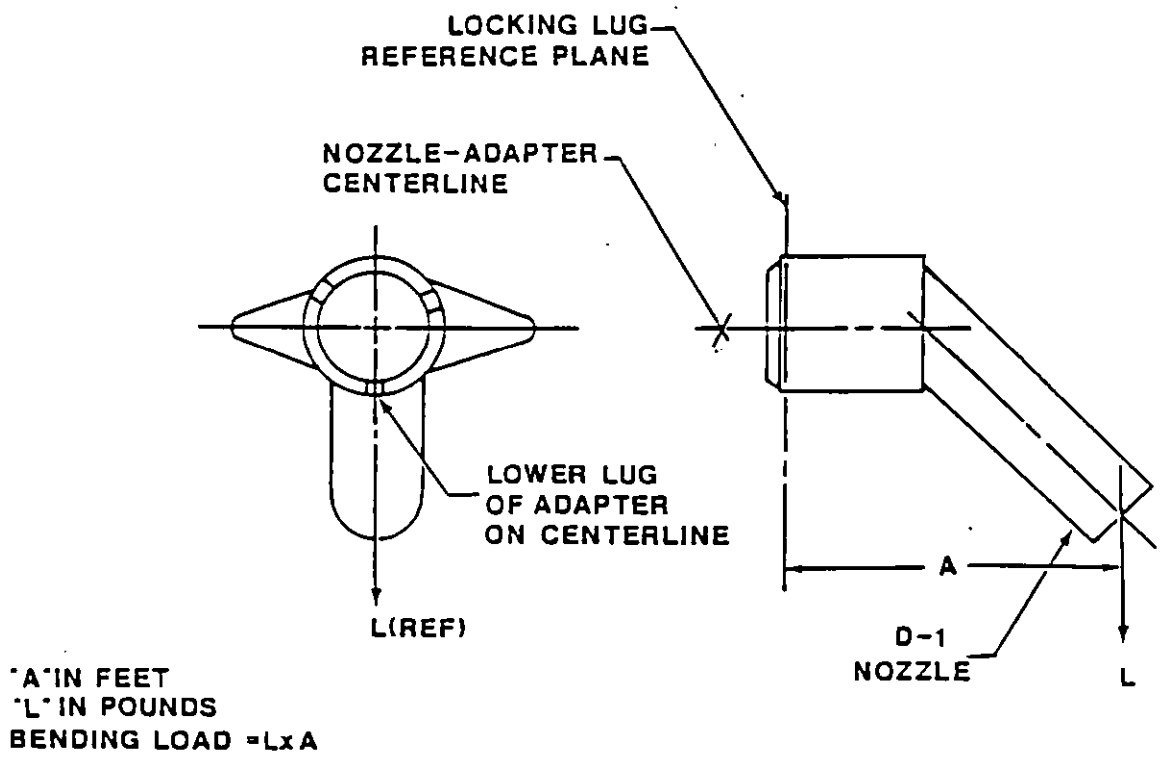


FIGURE 2. Strength test (bending).

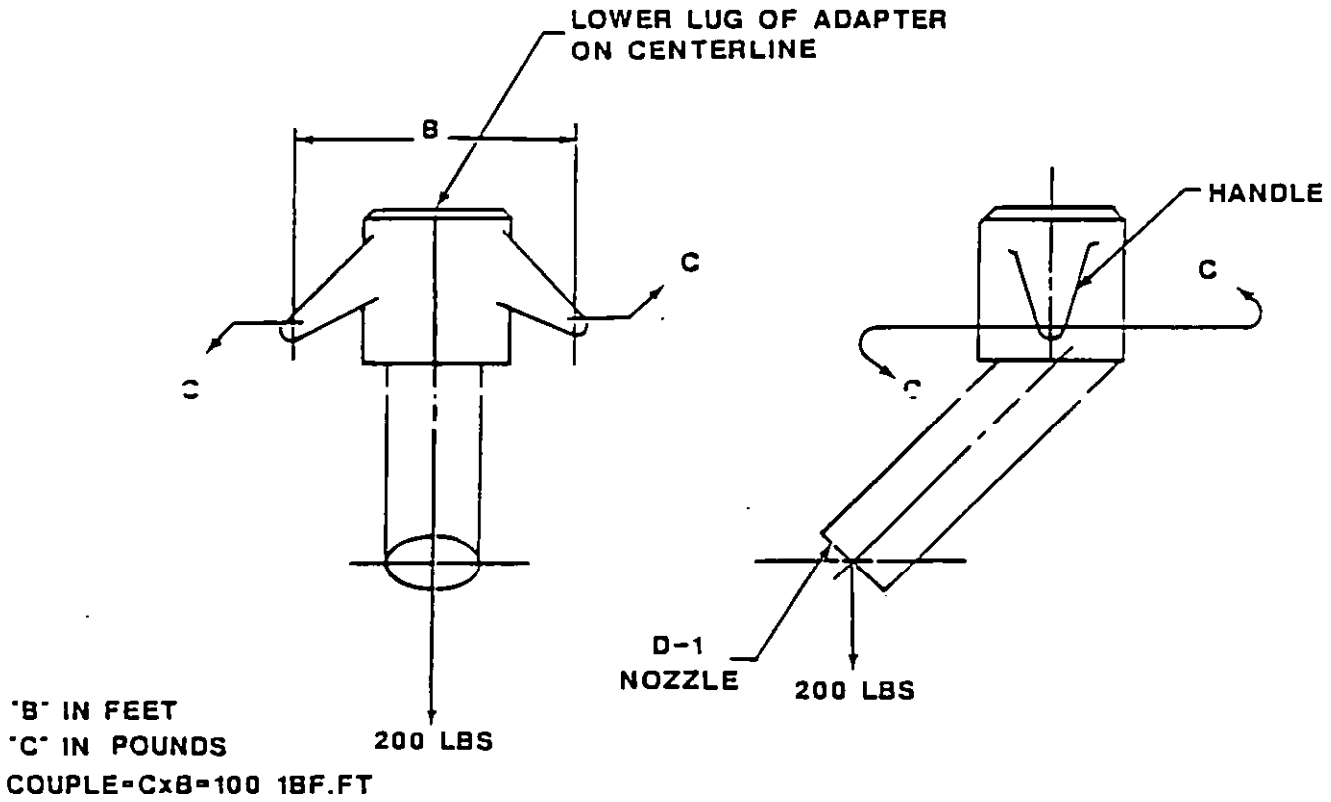
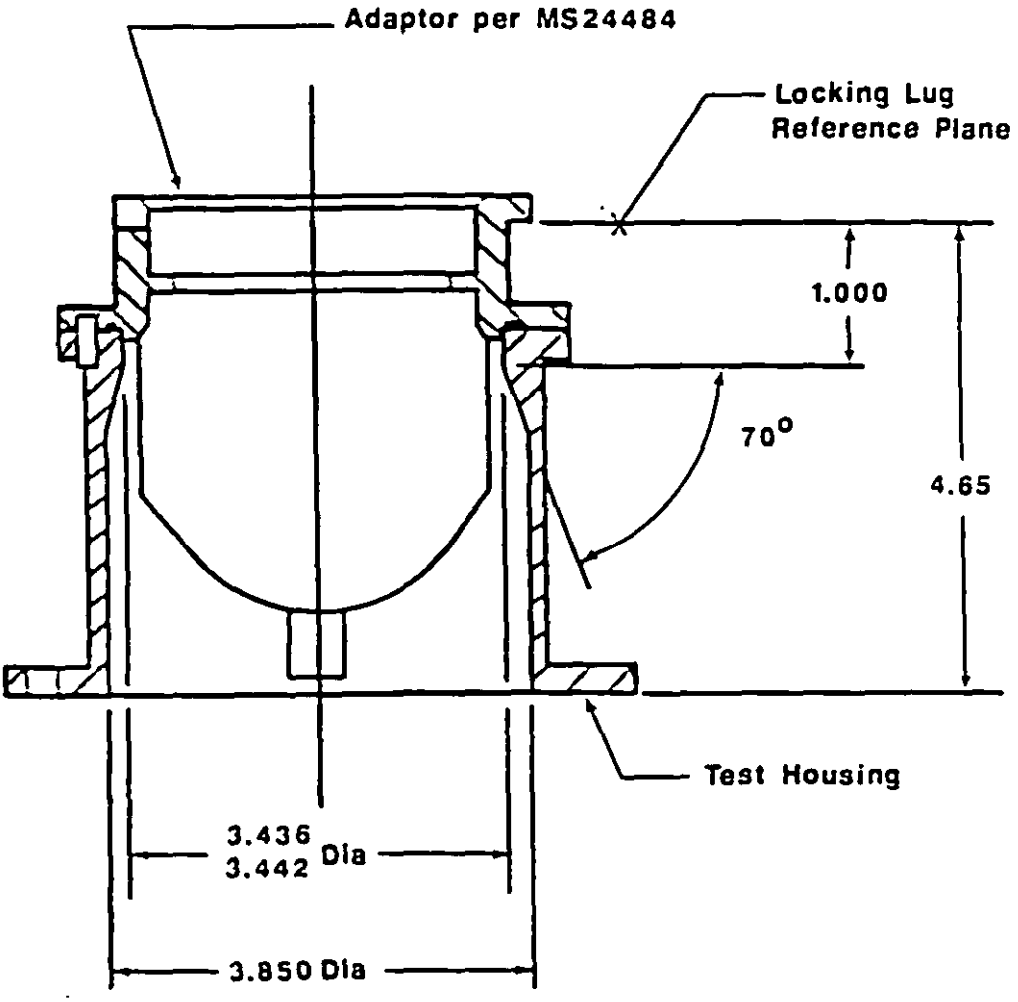


FIGURE 3. Strength test (combined loading).

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UNLESS OTHERWISE SPECIFIED
Dimensions are in inches.
Tolerances: .XX = ± .03
.XXX = ± .010
Angles = ± 1.0°

FIGURE 4. Test housing.

X-4846

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-N-53097(ME)	2. DOCUMENT DATE (YYMMDD) 900207

3. DOCUMENT TITLE: Nozzle, Pressure Fuel Servicing, Locking, Type D-1, Nominal 2-1/2-Inch Diameter, Arctic Service

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (703) 664-5717 (2) AUTOVON: 354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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